



COURSE SYLLABUS						
Cours e Code	COMP 013	Course Title	Human Computer Interaction		Course Credit	3
Course Description		Human Computer Interaction (HCI), serves as a vital component of the computing curriculum, bridging technology and human behavior to ensure that digital systems are intuitive, accessible, and efficient. The course nurtures students’ ability to design and evaluate interactive systems that respond to real-world user needs through a combination of analytical thinking, creativity, and technical proficiency. By integrating principles from computer science, cognitive psychology, and design, students develop the competence to build user-centered solutions that enhance productivity and promote inclusivity. Ultimately, HCI strengthens the learner’s understanding of how people interact with technology—preparing them to contribute to innovative, ethical, and sustainable system development in various professional computing fields.				
Pre-Requisites		COMP 002	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				





**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.





	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING	Based on CMO	QS WORLD UNIVERSITY RANKINGS 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia THE WORLD UNIVERSITY RANKINGS OF INNOVATION Rank 21 GENERATIVE AI APPLICATION Rank 39 STUDENT SUPPORT ENGAGEMENT Rank 50 SYMBOL/ PROMOTION	Alignment to ILOs <table><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>Employability Skills Learning</td><td>Teaching & Learning</td><td>Research & Innovation</td><td>Responsibility & Sustainability</td><td>Leadership & Governance</td><td>Discretion (SND)</td><td></td><td></td><td></td></tr><tr><td>Internationalization</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Times Higher Education Impact Rankings 2024</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>OF BUTS at 1501+</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	1	2	3	4	5	6	7	8	9	Employability Skills Learning	Teaching & Learning	Research & Innovation	Responsibility & Sustainability	Leadership & Governance	Discretion (SND)				Internationalization									Times Higher Education Impact Rankings 2024									OF BUTS at 1501+								
	1	2	3	4	5	6	7	8	9																																							
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Times Higher Education Impact Rankings 2024																																																
OF BUTS at 1501+																																																
	Apply knowledge of computing, science, and mathematics appropriate to the discipline																																															



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OUTCOMES (PLO)	Analyze complex problems, and identify and define the computing requirements appropriate to its solution												
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.												
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.												

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2025
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92nd in Southeast Asia
541-560 in Asia



Rank 21
GENERATIVE AI
APPLICATION

Rank 39
STUDENT SUPPORT
ENGAGEMENT

Rank 50
SYMBOL/
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	Integrate IT-based solutions into the user environment effectively.												
	Assist in the creation of an effective IT project plan.												
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession												
	Understand best practices and standards and their applications.												
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.												
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.												
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.												
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.												
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Explain the fundamental concepts, evolution, and interdisciplinary nature of Human-Computer Interaction (HCI) by identifying its core components—human, computer, and interaction—and describing how psychology, design, and computing principles integrate to enhance user experience.								
	Analyze human cognitive, physical, and perceptual characteristics that affect system usability and user experience, demonstrating an understanding of how human limitations and strengths influence								

WORLD UNIVERSITY RANKINGS 2025
7th in the Philippines
92nd in Southeast Asia
41-560 in Asia

QS STARS
5★
- Employability
- Online Learning
- Internationalization

WURI
The WORLD UNIVERSITY RANKINGS for INNOVATION

Impact Rankings 2024
DEBUTS AT 1501+

Social Responsibility
- Teaching
- Academic Development
- Specialized Criteria (Behavior of Science in Nutrition and Dietetics (BSND))



	interface design decisions.											
	Apply usability principles and interaction design guidelines to create user-friendly interfaces that emphasize accessibility, efficiency, learnability, and satisfaction according to industry standards and best practices.											
	Design and develop low- and high-fidelity prototypes using suitable tools (e.g., Figma, Adobe XD, or similar) that translate user requirements into interactive and aesthetically coherent interface layouts.											
	Conduct usability testing and evaluation through observation, heuristic analysis, and user feedback to identify usability issues and propose design improvements based on test results and iterative refinement											
	Integrate User-Centered Design (UCD) and accessibility principles in the design process, ensuring inclusivity for diverse user groups, including individuals with disabilities or special needs.											
	Critically assess emerging technologies and interaction paradigms such as augmented reality (AR), virtual reality (VR), artificial intelligence (AI), and multimodal interfaces, and evaluate their impact on human interaction and design practices.											
	Develop and present a complete HCI-based capstone project that demonstrates the integration of research, design, development, and evaluation processes—adhering to ethical, social, and professional standards in technology design.											





OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignmen t to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	- Understand the scope, importance, and evolution of HCI. - Identify key components of the HCI discipline.	- Introduction to HCI: Definition, Goals, History, and Importance. - The Human, the Computer, and the Interaction.	Lecture and discussion on HCI foundations; group discussion on components of interaction.	Explore real-world examples of user interfaces (e.g., mobile apps or websites) and identify HCI concepts applied in their design. Create a mind map showing the	Short quiz on HCI concepts and oral recitation.	Online quiz and reflection post; submission of digital mind map.				



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				three HCI components.					
3	Analyze human factors affecting user interaction.	Human Capabilities: Perception, Cognition, and Physical Interaction.	Demonstration and discussion on human limitations.	Research article summary on cognitive models in HCI.	Participation and discussion.	Online forum discussion			
4	Describe usability principles and goals of effective design.	Usability Goals, Measures, and Principles.	Case study analysis on usable interfaces.	Explore sample UI/UX applications and evaluate usability.	Quiz on usability metrics.	Online usability evaluation form.			
5	Apply design principles in interface design	Design Principles: Consistency, Feedback, Visibility, Affordance.	Workshop on interface design patterns.	Create a visual slide presentation on design principles.	Output check of design drafts.	Online submission of presentation.			
6	Explain design process models and their applications.	Interaction Design Process and Prototyping Models.	Lecture and group brainstorming.	Watch demo video on design process stages.	Group activity report.	Reflective online journal.			
7	Use storyboards and scenarios in design conceptualization.	Task Analysis, Scenarios, and Storyboards.	Hands-on activity: Create storyboards.	Online collaboration tool use (Miro/Figma).	Presentation of storyboard.	Upload of digital storyboard.			





8	- Demonstrate low-fidelity prototyping skills. - Design a high-fidelity prototype with interactive features.	- Paper Prototyping and Wireframing. - High-Fidelity Prototyping Tools (Figma, Adobe XD).	Workshop on sketching interfaces; guided software lab session.	Watch online tutorial and practice with Figma; continue project prototype remotely.	Face-to-face evaluation of sketches; prototype demonstration.	Screenshot submission of digital wireframe; online peer feedback form.				
9	MIDTERM WRITTEN EXAMINATION									
10	Conduct usability testing for a designed interface.	Usability Testing: Methods, Tools, and Evaluation.	Mock usability test simulation.	Record online usability test session.	Observation checklist.	Submission of usability report.				
11	Apply user-centered design (UCD) in prototype improvement.	User-Centered Design and Iterative Development.	Group discussion and design refinement.	Online collaboration and feedback exchange.	Revised prototype check.	Online submission of final prototype.				
12	Understand accessibility and inclusivity in design.	Accessibility Standards and Inclusive Design.	Lecture with examples of accessible designs.	Online research on accessibility standards (WCAG).	Quiz on accessibility standards.	Online article critique.				



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13	Apply evaluation methods for interactive systems.	Heuristic Evaluation and Cognitive Walkthroughs.	Workshop: Conduct heuristic evaluation.	Peer evaluation via online forms.	Group output presentation.	Upload evaluation checklist.				
14	Recognize emerging trends in HCI technologies.	Emerging Trends: AR/VR, AI Interfaces, Multimodal Interaction.	Lecture and demo of current tech tools.	Watch online tech demos and create reflections.	Participation and Q&A.	Online quiz and reflection paper.				
15	Apply ethics and professional standards in HCI.	Ethics, Privacy, and Social Implications in Design.	Case study discussion.	Analyze ethical dilemmas online.	Group report on case study.	Online discussion forum				
16	Integrate learned concepts in a capstone design project.	Capstone Project: Planning and Design.	Consultation and prototype development.	Online consultation sessions.	Prototype progress evaluation.	Submit digital prototype.				
17	- Present and defend final HCI project. - Reflect on overall HCI learning journey.	- Project Presentation and Defense. - Course Review, Reflection, and Future Trends.	Face-to-face presentation and defense; Synthesis and feedback session.	Prepare online documentation; Submit online reflection portfolio.	Final oral defense; Reflection sharing session.	Upload final report and prototype; Final reflection portfolio submission.				

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541-560 in Asia

WURI
The WORLD UNIVERSITY RANKINGS for INNOVATION

QS STARS
★★★★★
- Employability
- Online Learning
- Internationalization

Times Higher Education Impact Ranking
DEBUTS AT 150th



18	FINAL WRITTEN EXAMINATION



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PROMOTION



5 stars

Employability
Online Learning
Internationalization



4 stars

Teaching
Academic
Development
Inclusiveness



3 stars

Social Responsibility
Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)



Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+

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		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies ● Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		● Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: ● Figma. (n.d.). What is human-computer interaction (HCI). Figma Resource Library. https://www.figma.com/resource-library/human-computer-interaction/ ● Interaction Design Foundation. (n.d.). Interaction design. https://www.interaction-design.org/literature/topics/interaction-design ● Computer Science Field Guide. (n.d.). Human computer interaction. https://www.csfieldguide.org.nz/en/chapters/human-computer-interaction/ ● BotPenguin. (n.d.). Human-computer interaction (HCI): Key principles. https://botpenguin.com/glossary/human-computer-interaction ● HCI Studies. (n.d.). HCI studies. https://hci-studies.org/										





CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.

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SYMBOL/ PROMOTION

QS STARS

★★★★★

Employability
Online Learning
Internationalization

★★★★★

Teaching
Academic Development
Inclusiveness

★★★★★

Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education Impact Rankings 2024

DEBUTS AT 1501+



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is “INC” grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don’t tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student’s own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original





creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a



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PROMOTION



5 stars

Employability
Online Learning



Teaching
Academic
Development
Inclusiveness



Social Responsibility
Specialist Criteria: Bachelor
of Science in Nutrition and
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Times Higher Education
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DEBUTS AT 1501+



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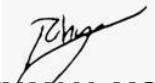


Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Exercises/Activities		Exercises/Activities	
Summative Quizzes		Summative Quizzes	
Recitation and Attendance		Recitation and Attendance	
Reflection Paper		Reflection Paper	
Midterm Examinations	30%	Final Examinations	
	100%		30%
			100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.		
COURSE REQUIREMENT/S	GRADING SYSTEM		
A team interface project with reports and presentation.	Final Grade = (Midterm Grade + Final Term Grade) / 2 QS WORLD UNIVERSITY RANKINGS 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia WURI The WORLD UNIVERSITY RANKINGS for INNOVATION Rank 21 Rank 39 Rank 50 GENERATIVE AI APPLICATION STUDENT SUPPORT ENGAGEMENT SYMBOL/ PROMOTION QS STARS ★★★★★ ★★★★★ ★★★★★ Employability Online Learning Teaching Academic Development Social Responsibility Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND) Internationalization Times Higher Education Impact Rankings 2024 PERUTS AT 150+		



CLASS INFORMATION	FACULTY INFORMATION
Course, Year and Section: BSIT 2-1	Name of Faculty: Prof. John Dustin Santos
Day and Time: Friday (4:00 pm – 7:00 pm)	Consultation Time:
Room: TGA308	Office Tel. No./ Mobile Phone No.
School Year: 2025-2026 Semester: 2 nd Semester	E-Mail Address:

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	 DR. RHYAN V. MOLINAR Director	DR. ARMANDO A. TORRES Vice President for Campuses

